SECTION A [40 MARKS]

ANSWER **ALL** QUESTIONS

Question 1 **[25]**

a) **Directions: For each question, there are four alternatives A, B, C and D. Choose the correct alternative and circle it. Do not circle more than ONE alternative. If there are more than one choice circled, NO score will be awarded.**

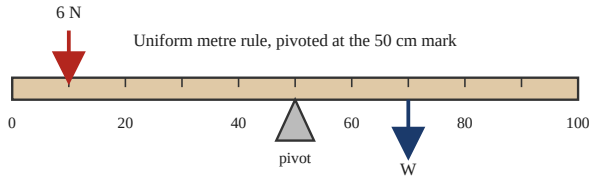
i. The SI unit of the moment of a force is

A N
B N m
C N/m
D J/s

ii. A body is in stable equilibrium if, when it is slightly displaced, its centre of gravity

A rises and the body returns to its original position.
B falls and the body moves further away.
C stays at the same height and the body stays in the new position.
D moves outside the base and the body topples.

iii. The uniform metre rule in the figure is balanced at the 50 cm mark by a 6 N force at the 10 cm mark. What is the weight W hung at the 70 cm mark?



A 4 N
B 6 N
C 9 N
D 12 N

iv. Which of the following objects has the greatest stability?

A A tall, narrow cylinder standing on its end
B A short cone with a wide base
C A cone balanced on its tip
D A tall box with a narrow base

v. If the distance between two masses is halved, the gravitational force between them becomes

A half
B double
C four times
D one quarter

vi. A stone is dropped from a cliff. Neglecting air resistance, its speed after 3 s is ($g = 10\ m/s^2$)

A 10 m/s
B 20 m/s
C 30 m/s
D 45 m/s

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vii. A parachutist falls at terminal velocity. At this stage

A the weight is greater than the air resistance.
B the weight is equal to the air resistance.
C the air resistance is zero.
D the acceleration is 10 m/s².

viii. A force of 50 N moves an object 4 m in the direction of the force. The work done is

A 12.5 J
B 54 J
C 100 J
D 200 J

ix. A crane lifts a 500 kg load through 10 m in 20 s. The power developed is (g = 10 m/s²)

A 250 W
B 2500 W
C 5000 W
D 50 000 W

x. A machine takes in 500 J of energy and gives out 400 J of useful energy. Its efficiency is

A 20 %
B 80 %
C 100 %
D 125 %

xi. A force of 200 N acts on an area of 0.5 m². The pressure is

A 100 Pa
B 250 Pa
C 400 Pa
D 1000 Pa

xii. In a hydraulic machine a force of 20 N acts on a piston of area 2 cm². The large piston has an area of 40 cm². The force on the large piston is

A 20 N
B 40 N
C 200 N
D 400 N

xiii. The pressure at a point in a liquid depends on

A the shape of the container.
B the depth and the density of the liquid.
C the area of the surface of the liquid.
D the volume of the liquid only.

xiv. How much heat is needed to raise the temperature of 3 kg of water by 10 °C? (c = 4200 J/kg°C)

A 12 600 J
B 42 000 J
C 126 000 J
D 420 000 J

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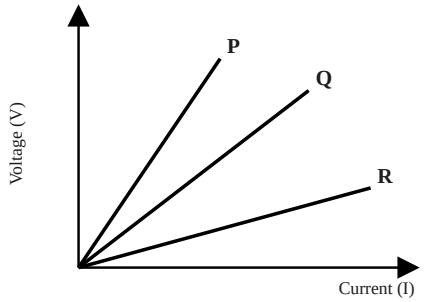
xv. Equal amounts of heat are supplied to equal masses of four liquids. Which liquid shows the smallest rise in temperature?

A Liquid A, c = 2100 J/kg°C
B Liquid B, c = 2400 J/kg°C
C Liquid C, c = 3400 J/kg°C
D Liquid D, c = 4200 J/kg°C

xvi. While ice is melting at 0 °C, the heat supplied

A raises the temperature of the ice.
B raises the temperature of the water.
C changes the state without a change in temperature.
D is lost to the surroundings.

xvii. The graph shows the V–I relationship for three conductors P, Q and R. Which conductor has the greatest resistance?



A P
B Q
C R
D All have the same resistance

xviii. A 12 Ω resistor is connected across a 36 V supply. The current through it is

A 0.33 A
B 3 A
C 24 A
D 432 A

xix. The heat produced in a 5 Ω resistor carrying a current of 2 A for 10 s is

A 20 J
B 100 J
C 200 J
D 400 J

xx. Fleming's left-hand rule is used to find the direction of

A the induced current in a generator.
B the force on a current-carrying conductor in a magnetic field.
C the magnetic field around a straight wire.
D the heat produced in a conductor.

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xxi. A transformer has 100 turns in the primary and 500 turns in the secondary. If the input voltage is 20 V, the output voltage is

A 4 V
B 20 V
C 100 V
D 2500 V

xxii. Which of the following electromagnetic waves has the shortest wavelength?

A Microwaves
B Infrared
C X-rays
D Gamma rays

xxiii. Which of the following is a longitudinal wave?

A Light
B Sound
C X-ray
D Radio wave

xxiv. An alpha particle consists of

A 1 proton and 1 neutron.
B 2 protons and 2 neutrons.
C 1 electron.
D a high-energy electromagnetic wave.

xxv. When uranium-238 ($^{238}_{92}U$) emits an alpha particle, the daughter nucleus is

A $^{234}_{90}Th$
B $^{234}_{92}U$
C $^{238}_{93}Np$
D $^{236}_{90}Th$

b) Fill in the blanks by writing suitable word/s. **[5]**

i.	The turning effect of a force about a point is called the _____.	
ii.	At terminal velocity the net force acting on a falling body is _____.	
iii.	Heat always flows from a body at a higher temperature to a body at a _____ temperature.	
iv.	The direction of the induced current in a conductor is given by Fleming's _____ rule.	
v.	The radiation which has the highest penetrating power is _____.	

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c) State whether the following statements are **TRUE** or **FALSE**. **[5]**

i.	A couple produces a net force on a body.		
ii.	The pressure at a point in a liquid acts in all directions.		
iii.	The resistance of a wire increases when its length is increased.		
iv.	Radio waves travel faster than gamma rays in a vacuum.		
v.	The red shift of light from distant galaxies shows that the universe is expanding.		

d) Match each item under **Column A** with the most appropriate item in **Column B**. Write the correct alphabet in the space provided under the '**ANSWER**' column. **[5]**

Answer	Column A	Column B	
i.	i. Unit of power	a. latent heat	
ii.	ii. Device that changes an alternating voltage	b. watt	
iii.	iii. Heat absorbed in melting at constant temperature	c. beta rays	
iv.	iv. A vehicle that moves over the surface of Mars	d. rover	
v.	v. Ray deflected towards the positive plate	e. transformer	
		f. alpha rays	
		g. joule	

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SECTION B [60 MARKS]

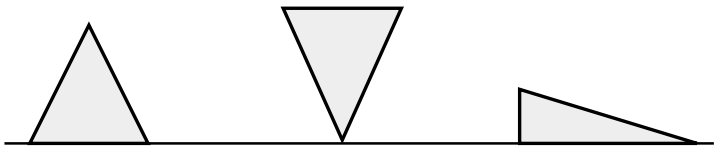
ATTEMPT **ANY SIX** QUESTIONS

Question 2

a) A uniform plank 4 m long is pivoted at its centre. A boy weighing 300 N sits 1.5 m from the pivot. Where should a girl weighing 450 N sit to balance the plank? **[2]**

b) State TWO conditions for a body to be in equilibrium. **[1]**

c) Three pictures of the same cone resting on a table are shown. Name the type of equilibrium of each and give a reason. **[2]**



d) A table lamp with a wide, heavy base is difficult to topple. Explain. **[1]**

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e) Which produces the greater moment about the pivot: 20 N acting at 0.30 m or 15 N acting at 0.50 m? Show your calculation. **[2]**

f) Turning a tap is an example of a couple. Why? **[2]**

Question 3

a) A ball is dropped from rest from the top of a tower and takes 4 s to reach the ground. Take g = 10 m/s². Calculate (i) the velocity of the ball just before it hits the ground and (ii) the height of the tower. **[2]**

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b) Explain why a feather and a coin fall at the same rate in a vacuum but not in air. **[2]**

c) Calculate the gravitational force between a 5 kg mass and an 8 kg mass placed 0.4 m apart. ($G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}) **[2]**

d) State the law of conservation of energy. **[1]**

e) A girl of mass 60 kg climbs a flight of stairs 3 m high in 6 s. Calculate the power she develops. ($g = 10$ m/s^2) **[2]**

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f) Give ONE way in which a household can improve its energy efficiency. **[1]**

Question 4

a) In a hydraulic jack the small piston has an area of 4 cm^2 and a force of 50 N is applied to it. The large piston has an area of 80 cm^2. Calculate the force on the large piston and name the principle used. **[2]**

b) A sharp knife cuts better than a blunt one. Explain. **[1]**

c) A rectangular tank with a base 2 m × 1.5 m contains water to a depth of 2 m. Calculate (i) the pressure at the bottom and (ii) the force on the base. (density of water = 1000 kg/m^3, g = 10 m/s^2) **[2]**

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d) Calculate the heat needed to raise the temperature of 0.5 kg of water from 20 °C to 70 °C. (c = 4200 J/kg°C) **[2]**

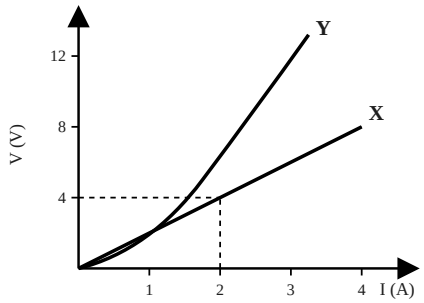
e) 1 kg of water at 90 °C is mixed with 2 kg of water at 15 °C in an insulated container. Calculate the final temperature. **[2]**

f) Why is water used as a coolant in car radiators? **[1]**

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Question 5

a) The graph shows the V–I graphs for two wires X and Y. Which wire obeys Ohm's law? Use the graph to calculate the resistance of that wire. **[2]**



b) A 60 W lamp is connected to a 240 V supply. Calculate (i) the current through the lamp and (ii) its resistance. **[2]**

c) The filament of a lamp glows but the connecting wires stay cool although the same current flows through both. Explain. **[1]**

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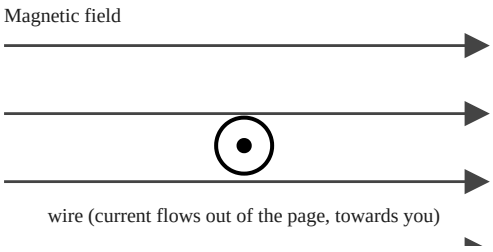
d) Calculate the heat produced in a 20 Ω resistor carrying a current of 3 A for 5 s. **[2]**

e) State TWO factors which affect the resistance of a wire. **[1]**

f) Explain why electricity is transmitted over long distances at a very high voltage. **[2]**

Question 6

a) A straight wire carrying a current out of the page lies in a magnetic field directed from left to right as shown. State the direction of the force on the wire and name the rule you used. **[2]**



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b) Name ONE way of increasing the emf produced by a generator. **[1]**

c) A transformer has 400 turns in the primary coil and 100 turns in the secondary coil. The input is 240 V and 0.5 A. Assuming the transformer is 100 % efficient, calculate (i) the output voltage and (ii) the output current. **[3]**

d) Why does a transformer not work with a direct current (DC) supply? **[1]**

e) State TWO differences between an AC generator and a DC generator. **[2]**

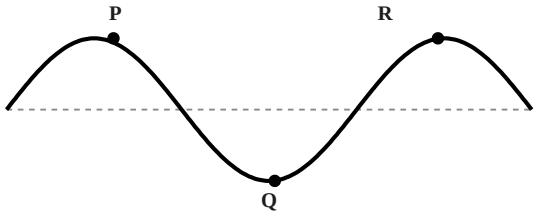
f) The core of a transformer is made of laminated iron sheets. Why? **[1]**

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Question 7

a) An FM radio station broadcasts at a frequency of 100 MHz. Calculate the wavelength of the wave. (speed of radio waves = 3×10^8 m/s) **[2]**

b) The figure shows a wave. Identify the distance between P and R and the distance from Q to the dotted line. Is it a transverse or a longitudinal wave? **[2]**



c) Why is infrared radiation used in a TV remote control? State ONE limitation. **[2]**

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d) State ONE harmful effect of ultraviolet radiation on the human body. **[1]**

e) State ONE advantage of digital signals over analogue signals. **[1]**

f) A person in a valley behind a hill can hear a radio broadcast but cannot see the transmitter. Explain. **[2]**

Question 8

a) Complete the equation for the decay of radium: $^{226}_{88}Ra \rightarrow X + {}^{4}_{2}He$ **[1]**

b) A lead nucleus $^{214}_{82}Pb$ emits a beta particle. Write the nuclear equation. **[2]**

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c) Name the radiation with (i) a charge of +2 and (ii) no charge and the greatest penetrating power. **[2]**

d) State ONE safety precaution when handling a radioactive source in a laboratory. **[1]**

e) Why is gamma radiation used to sterilise surgical instruments? **[1]**

f) Differentiate between an orbiter and a rover. **[2]**

g) What does the cosmological red shift tell us about the universe? **[1]**

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Question 9

a) Suggest TWO features that would make a school hostel in a cold place energy efficient. **[2]**

b) A car of mass 1000 kg moves at 20 m/s. Calculate its kinetic energy. How much work must the brakes do to stop it? **[2]**

c) Should Bhutan install more solar power plants? Justify with TWO reasons. **[2]**

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d) A boy of mass 40 kg stands on both feet. The total area of contact is 0.04 m^2. Calculate the pressure he exerts on the ground. ($g = 10\ m/s^2$) **[2]**

e) A tall bookshelf is usually fixed to the wall. Why is this necessary? **[1]**

f) State ONE advantage and ONE limitation of a flyby mission such as Voyager 2. **[1]**

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